

01 — Engineering Identity

Purpose

This chapter defines the long-term engineering identity, career direction, and professional compass that guide every decision within Mohammed road.

Engineering Identity

I build intelligent systems that solve real-world engineering problems by combining AI, computer vision, robotics, and trustworthy machine learning/DL.

Professional Goal

Become an Applied AI & Robotics R&D Engineer specializing in Intelligent Physical Systems and Trustworthy AI while completing my PhD and building financial stability for my future family.

Career Direction

- Engineering Identity
- Professional Goal
- Technical Domain: Intelligent Physical Systems
- Technical Specialization: Trustworthy AI for Intelligent Physical Systems

Current Research Direction

Develop trustworthy deep learning methods for computer vision, currently focused on image classification, while building foundations for future research in trustworthy perception and intelligent physical systems.

Long-Term Research Interest

Develop trustworthy AI methods for intelligent physical systems, including perception, localization, navigation, and related topics as research evolves.

Target Employer Blueprint

Target organizations build intelligent physical systems where AI, robotics, sensing, and computer vision are core technologies.

Representative sectors:

- Robotics
- Medical Devices
- Industrial Automation
- Autonomous Systems
- Advanced Manufacturing
- Applied AI R&D

Success Principles

1. Build flagship projects.
2. Produce research with engineering value.
3. Maintain evidence-driven portfolio.
4. Continuously improve engineering competencies.

SO:

My professional center:

Applied AI & Robotics R&D

My Future engineering domain:

Intelligent Physical Systems

My major technical pillars will be:

AI / Machine Learning / Deep Learning

Computer Vision

Robotics & Physical Systems

Mathematics / Optimization / ML Foundations

My additional specialization will be:

Trustworthy Machine Learning / Deep Learning

My current strongest evidence:

Deep Learning + Computer Vision + ML research

My engineering foundation

Mechatronics + sensing + embedded systems + robotics + physical system design

And then different applications can branch from there:

Computer Vision

Robotics

Medical Devices

Industrial Automation

Autonomous Systems

02 — Engineering Foundation

Purpose

This chapter establishes the conceptual foundation of Mohammed life by defining the permanent engineering architecture used throughout the handbook.

Core Definitions

Competency: A long-term engineering ability that remains stable despite changing technologies.

Capability: A practical engineering ability that implements a competency.

Knowledge Area: The theoretical knowledge required to develop one or more capabilities.

Project: An engineering activity used to demonstrate capabilities.

Evidence: Objective proof that a capability has been demonstrated.

Canonical Engineering Flow

Competency – Capability – Knowledge – Project – Evidence

Engineering Competency Framework

- Mathematics
- Programming & Software Foundations
- AI & Machine Learning
- Computer Vision
- Robotics & Intelligent Physical Systems
- Trustworthy AI
- Professional AI Engineering

Evidence Philosophy

Competence is measured through evidence rather than certificates. Preferred evidence includes flagship projects, publications, public repositories, technical reports, benchmarks, demonstrations, and industrial experience.

Design Principles

5. Competencies are technology-independent.
6. Capabilities are observable and measurable.
7. Projects generate evidence.

8. Evidence drives career progression.
9. Continuous improvement is iterative.